THE KNOWLEDGE UNIVERSITY

IT4010 – Research Project - 2024

Topic Assessment Form

Project ID: **R25-068**

1. Topic (12 words max)

Ayurvedic Medicine Identification and Health Management App

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Problem Statement: Despite the increasing interest in holistic health and alternative therapies, the management of skin conditions remains a significant challenge. Common skin issues such as psoriasis, vitiligo, and other inflammatory disorders affect millions, leading to both physical discomfort and emotional distress. Current treatment options often rely heavily on chemical products, which may cause adverse effects and contribute to environmental degradation. Additionally, patients frequently seek natural alternatives that promote well-being without the risks associated with conventional medications. However, the lack of accessible information and guidance on Ayurvedic remedies limits the effective integration of these practices into skin health management.

Yu, K., Beauchamp, J. E., & Niu, M. (2019). Artificial intelligence in dermatology: Current applications and future directions. Journal of Dermatological Science, 94(1), 38-44.

Kumar, S., et al. (2021). Ethnopharmacological approaches to the treatment of skin diseases in India: A review. Journal of Ethnopharmacology, 267, 113591.

Research Gap: The existing literature and technological solutions do not provide a comprehensive approach to skin health management that leverages advancements in artificial intelligence and traditional Ayurvedic medicine. Specifically, there is a notable absence of systems that utilize AI for accurate skin condition identification and personalized treatment recommendations based on Ayurvedic principles. Current diagnostic methods often depend on subjective assessments by healthcare professionals, resulting in variability and inconsistency in treatment plans. Furthermore, there is a lack of resources that educate users on the medicinal properties of Ayurvedic plants and their applications for skin health. Addressing these gaps requires innovative research to develop integrated systems that combine AI technology with Ayurvedic practices, ultimately promoting a holistic and effective approach to skin health management.

The Ayurvedic mobile app provides a comprehensive, technology-driven solution for managing autoimmune and skin disorders, identifying medicinal plants, analyzing oral health, and recognizing animal harm. By integrating advanced AI technology and Ayurvedic principles, the app delivers personalized health insights and recommendations.

Autoimmune & Skin Disorders: The app offers Ayurvedic treatments for conditions like Psoriasis, Vitiligo, Eczema, and Acne, providing tailored herbal remedies, therapies (such as Panchakarma), and lifestyle suggestions based on the severity of the condition (mild, moderate, or severe).

Medicinal Plant Identification: AI-powered plant recognition identifies various medicinal plants and links them to their traditional health benefits. The app offers preparation methods, dosages, and detailed information about how these plants can treat various ailments, from skin conditions to digestive disorders.

Oral Health Analysis (Tongue & Skin Texture): The app includes features for analyzing oral health. Users can upload an image of their tongue, and the AI detects key indicators such as cracks, color, coating, and texture, which provide insights into their digestive health, toxicity levels, and dosha imbalances. This component helps users monitor oral health and offers Ayurvedic treatment suggestions to address imbalances.

Animal Harm Recognition: Users can upload photos or videos of animals or insects causing harm. The AI analyzes these images to identify injuries, diseases, or harmful species, providing Ayurvedic remedies for treatment and first-aid. The app also connects users with local Ayurvedic practitioners for further consultation and care.

This integrated system combines AI-driven technology with Ayurvedic healing, providing a comprehensive solution for managing health, plant-based treatments, and animal welfare, while promoting sustainable and holistic wellness. me

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Specialized Domain Expertise, Knowledge, and Data Requirements

The development of the Ayurvedic Medicine Identification and Health Management App requires expertise from multiple specialized domains to ensure accurate diagnostics, recommendations, and treatment guidance.

1. Specialized Domain Expertise:

Ayurvedic Medicine Specialists: Experts in Ayurvedic treatments, medicinal plants, and traditional healing methods to create a comprehensive knowledge base.

Botanists and Herbalists: For accurate plant identification and understanding of medicinal properties.

Dermatologists and Ayurvedic Practitioners: To validate skin disease analysis and corresponding Ayurvedic remedies.

Veterinarians and Animal Health Experts: To guide the development of the animal harm recognition module.

Data Scientists and Machine Learning Engineers: To design, train, and deploy AI models for image recognition and predictive analysis.

Mobile App Developers: For creating a user-friendly, secure, and scalable mobile application.

2. **Knowledge Requirements:**

Ayurvedic texts and research papers for disease classification and treatment protocols.

Botanical reference guides for plant taxonomy.

Medical literature on dermatological and oral health conditions.

Veterinary guidelines for animal first-aid and Ayurvedic treatment compatibility.

3. **Data Requirements:**

Image Datasets: High-quality, labeled images of medicinal plants, skin conditions, tongue types, and common animal injuries.

Ayurvedic Remedy Database: Recipes, dosages, preparation methods, and usage instructions.

Medical Case Studies: Real-world treatment data for AI model training and validation.

8. Objectives and Novelty

The objective of this research project is to develop an advanced AI-powered system that offers personalized solutions for skin health management, integrating Ayurvedic wisdom and modern dermatological practices. This system aims to address various skin conditions by providing accurate identification, treatment recommendations, and preventive measures through image analysis and a comprehensive knowledge database. By utilizing image recognition technology, the system will enable users to identify Ayurvedic medicinal plants and remedies for skin issues, facilitating improved skin health and overall well-being. The integration of user feedback mechanisms will ensure continuous learning and adaptation of the system, enhancing the effectiveness of personalized recommendations. Ultimately, this project seeks to promote holistic skin health management while minimizing reliance on chemical treatments and fostering sustainable practices in skin care.

Member Name	Sub Objective	Tasks	Novelty
Peiris. P.H.R (IT21584718)	To develop an AI-powered mobile application to identify Ayurvedic medicinal plants through image analysis, offering users detailed insights into their medicinal properties and applications.	• Capture and upload high-quality images of Ayurvedic plant parts such as leaves, flowers, or stems. • Train AI and machine learning models to analyze images and accurately identify plant species. • Integrate a database to provide detailed descriptions of the health benefits and medicinal uses of the identified plants. • Process analyzed data to generate specific recommendations for health	This objective introduces a novel application that bridges the gap between traditional Ayurvedic medicine and modern technology. The use of AI for accurate identification of Ayurvedic medicinal plants ensures accessibility even with partial or low-quality images. Unlike generic plant identification tools, this app focuses on Ayurvedic plants and their medicinal value, promoting the preservation and modernization of traditional health practices. By leveraging advanced image

		management and traditional remedies. • Design an intuitive mobile interface for capturing images, displaying results, and accessing information seamlessly.	analysis and offering a comprehensive medicinal database, it empowers users to integrate traditional remedies into contemporary health management.
Fernando W.M.S. K (IT21813184)	Develop a system that provides personalized answers and recommendations for skin-related issues, integrating knowledge from Ayurvedic treatments and modern skin care practices.	Knowledge Base Development: Compile a comprehensive database of skin conditions (e.g., psoriasis, vitiligo) and their treatments, including Ayurvedic remedies and modern practices • Ensure the database is continuously updated with the latest research and findings. • System Design and Development Create a user-friendly interface that allows users to easily interact and ask questions about skin conditions. Implement Natural Language Processing (NLP) to understand user inquiries	Provides tailored advice that considers Ayurvedic and modern treatment modalities, creating a holistic approach to skin care. Holistic Health Focus: Emphasizes sustainable and natural remedies, promoting skin health while minimizing reliance on chemical treatments, appealing to eco-conscious consumers.

		and provide relevant answers. • Integration of Treatment Suggestions Develop algorithms that analyze user input and recommend tailored treatments based on specific skin conditions and individual preferences. Include recommendations for Ayurvedic herbs, therapies, and lifestyle changes. • User Feedback Loop: Implement a feedback mechanism for users to report the effectiveness of the suggested treatments, enabling continuous improvement of the system. Utilize user input to refine the system's understanding and recommendations over time.	

		• Testing and Iteration: Conduct user testing to assess the system's performance and accuracy in providing information and recommendations. Iteratively improve the system based on user feedback and test result	
Huznadh M.N.M (IT21802812)	Develop an AI-driven system for animal harm recognition and treatment through Ayurvedic methods.	• Create a database of animal injuries and harmful insect species. • Train AI models for analyzing photos and videos of animals for injuries or abnormalities. • Provide detailed Ayurvedic treatment recommendations, including dosages and preparation methods for herbal remedies. • Develop connectivity features that link users to Ayurvedic veterinarians and hospitals for advanced care.	This system introduces an innovative way to integrate Ayurveda and AI, ensuring precise and efficient animal care. The novelty lies in combining modern AI capabilities with traditional Ayurvedic remedies to provide a holistic solution for animal care. The use of AI to identify harmful insects or conditions affecting animals is unique and expands the usability of Ayurvedic practices to veterinary health care. The addition of a geolocation feature for Ayurvedic hospital discovery enhances the app's utility, especially in rural or underserved areas.

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<td>Kaushalya H.G. B (IT21190520)</td>
<td>To develop an AI-powered application for oral care analysis using tongue imaging to detect health indicators such as digestive health, toxicity levels, and dosha imbalances, offering personalized Ayurvedic recommendations.</td>
<td>
Capture high-resolution tongue images to assess cracks, color (red, pale, or yellowish), coating, and texture.
Train AI models to analyze tongue features and correlate findings with Ayurvedic health indicators.
Identify key indicators such as digestive health, toxicity levels, and imbalances in Vata, Pitta, or Kapha doshas.
Provide tailored oral and dietary care recommendations based on the analysis.
Train machine learning models using datasets of annotated tongue images with Ayurvedic diagnostic insights.
Develop a database linking tongue characteristics to specific health indicators and personalized remedies.
</td>
<td>This application revolutionizes oral care by integrating AI-based tongue analysis with Ayurvedic diagnostics, offering insights beyond conventional oral health assessments. The ability to analyze subtle tongue characteristics such as texture, cracks, and color enables early detection of health imbalances tied to doshas. By providing personalized Ayurvedic recommendations for improving oral and systemic health, the app preserves traditional wisdom while making it accessible through modern technology. Its unique focus on tongue analysis positions it as a pioneering tool in holistic oral care.</td>
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		• Design a user-friendly interface for image capture, displaying analysis results, and providing recommendations.	

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Sanjeevi	Chandrasiri	Sanjeevi
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	√	No	

b) Does the proposed topic exhibit novelty?

Yes	√	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	√	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	√	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	

Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	**Signature**

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.